

WORK SHEET - 04

VECTOR & 3D

[HINTS & SOLUTIONS]

Q.1 (7, 2, 8)

Sol. Equation of line L is $\frac{x-1}{3} = \frac{y-2}{0} = \frac{z-0}{4} = \lambda$ (say)

$\therefore$ Any point on it is $(3\lambda + 1, 2, 4\lambda)$

Now, above point will satisfy $x - y + z = 13$, so $(3\lambda + 1) - 2 + 4\lambda = 13$

$\Rightarrow 7\lambda = 14 \Rightarrow \lambda = 2$

So, co-ordinates of Q is (7, 2, 8).

Q.2 4π

Sol. Let $\vec{a} = \hat{i} + \pi\hat{k}$; $\vec{b} = 3\hat{i} + \pi\hat{k}$ and $\vec{c} = -4\hat{i} + 2\hat{j} + \pi\hat{k}$

$$\begin{aligned} \therefore \text{Volume of parallelopiped} &= \left| \begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} \right| = \begin{vmatrix} 1 & 0 & \pi \\ 3 & 0 & \pi \\ -4 & 2 & \pi \end{vmatrix} \\ &= 1(0 - 2\pi) - 0 + \pi(6) = 6\pi - 2\pi = 4\pi. \end{aligned}$$

Q.3 $\frac{25\sqrt{61}}{36}$

Sol. The normal vector of plane P is parallel to vector $= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 0 & 1 \\ 0 & 3 & 2 \end{vmatrix} = \hat{i}(-3) - \hat{j}(4) + \hat{k}(6) = -3\hat{i} - 4\hat{j} + 6\hat{k}$

$\therefore$ Equation of plane P is $-3(x-0) - 4(y-1) + 6(z+1) = 0 \Rightarrow -3x - 4y + 6z + 10 = 0$

$$\Rightarrow -3x - 4y + 6z = -10 \text{ or } \frac{x}{10/3} + \frac{y}{10/4} + \frac{z}{-10/6} = 1$$

$$\begin{aligned} \therefore \text{Area of triangle ABC} &= \frac{1}{2} \sqrt{\left(\frac{100}{9}\right)\left(\frac{100}{16}\right) + \left(\frac{100}{16}\right)\left(\frac{100}{36}\right) + \left(\frac{100}{36}\right)\left(\frac{100}{9}\right)} \\ &= 50 \sqrt{\frac{1}{9 \times 16} + \frac{1}{16 \times 36} + \frac{1}{36 \times 9}} = \frac{50 \sqrt{36 + 9 + 16}}{3 \times 4 \times 6} = \frac{50\sqrt{61}}{72} = \frac{25\sqrt{61}}{36}. \end{aligned}$$

Q.4 $\frac{11}{2}$

$$\text{Sol. } V = \frac{1}{6} \begin{vmatrix} 8 & 1 & a^2 + 4a + 5 \\ -4 & 1 & 4\cot^2 \theta \\ -5 & -1 & \tan^2 \theta \end{vmatrix}$$

$$\begin{aligned}
 &= \frac{1}{6} (8 (\tan^2 \theta + 4 \cot^2 \theta) - 1(-4 \tan^2 \theta + 20 \cot^2 \theta) + (a^2 + 4a + 5)(4 + 5)) \\
 &= \frac{1}{6} (9(a^2 + 4a + 5) + 12 (\tan^2 \theta + \cot^2 \theta)) \\
 &= \frac{1}{6} (9(a+2)^2 + 1) + 12 (\tan^2 \theta + \cot^2 \theta) \\
 \therefore V_{\min} &= \frac{1}{6} (9 + 12 \times 2) = \frac{11}{3}.
 \end{aligned}$$

Q.5 1

Sol. We have
$$V = \begin{vmatrix} 1 & 1 & 0 \\ 1 & 2 \operatorname{cosec} \alpha & 1 \\ 0 & 1 & 2 \operatorname{cosec} \alpha \end{vmatrix} = 4 \operatorname{cosec}^2 \alpha - 1 - 2 \operatorname{cosec} \alpha$$

$$= 4 \left[\operatorname{cosec}^2 \alpha - \frac{1}{2} \operatorname{cosec} \alpha \right] - 1 = 4 \left(\operatorname{cosec} \alpha - \frac{1}{4} \right)^2 - \frac{5}{4}$$

$\therefore V_{\text{least}} \left(\alpha = \frac{\pi}{2} \right) = 4 \times \frac{9}{16} - \frac{5}{4} = \frac{4}{4} = 1.$

Q.6 8

Sol. Obviously both lines are parallel, point on 1st line is $(0, 2, -5)$ which must lie on 2nd line

$$\Rightarrow \frac{0-1}{2} = \frac{2-\lambda}{6} = \frac{-5-\mu}{4} = \lambda = 5, \mu = -3.$$

Q.7 9

Q.8 $2x + 17y - 7z = 33$

Q.9 7

Sol. We have

$$L_1: \frac{x-7}{-3} = \frac{y-6}{2} = \frac{z-2}{4} = \lambda$$

and $L_2: \frac{x-5}{2} = \frac{y-3}{1} = \frac{z-4}{3} = \mu$

Now,
$$\frac{2-3\lambda-2\mu}{2} = \frac{3+2\lambda-\mu}{-2} = \frac{-2+4\lambda-3\mu}{-1}$$

$\therefore 3\lambda + 2\mu - 2 = 3 + 2\lambda - \mu \Rightarrow \lambda + 3\mu = 5$ (1)

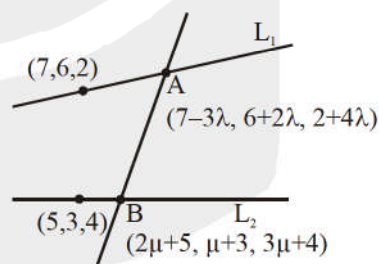
and $3\lambda + 2\mu - 2 = -4 + 8\lambda - 6\mu \Rightarrow 5\lambda - 8\mu = 2$ (2)

$\therefore$ On solving (1) and (2), we get

$$\lambda = 2, \mu = 1$$

So, $A \equiv (1, 10, 10)$ and $B \equiv (7, 4, 7)$

(i) $AB = \sqrt{(7-1)^2 + (4-10)^2 + (7-10)^2} = \sqrt{36 + 36 + 9} = \sqrt{81} = 9.$ Ans.



- (ii) Let equation of plane parallel to L_1 and containing L_2 be
 $a(x-5) + b(y-3) + c(z-4) = 0$ (3)

Now, $2a + b + 3c = 0$ and $-3a + b + 4c = 0$

$$\Rightarrow \frac{a}{-2} = \frac{b}{-17} = \frac{c}{7}$$

$\therefore$ From equation (3), we get

$$-2(x-5) - 17(y-3) + 7(z-4) = 0$$

or $2x + 17y - 7z = 33$. **Ans.**

- (iii) Volume of tetrahedron OPAB (where O is origin)

$$= \frac{1}{6} \left| \begin{vmatrix} \overrightarrow{OP} & \overrightarrow{OA} & \overrightarrow{OB} \end{vmatrix} \right| = \frac{1}{6} \left| \begin{vmatrix} 1 & 2 & 3 \\ 1 & 10 & 10 \\ 7 & 4 & 7 \end{vmatrix} \right| = \frac{1}{6} |42| = 7.$$

Q.10 $\vec{a} \cdot \vec{b} = \frac{-3}{5}$

$$[\vec{a} \vec{b} \vec{c}] = -4$$

$$[\vec{a} \vec{b} \vec{c}] = 4$$

Sol. $|\vec{a} + \vec{b}|^2 = |\vec{a} \times \vec{b}|^2$

$$a^2 + b^2 + 2\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \sin \theta$$

$$2 + 2 \cos \theta = \sin \theta$$

$$2 = \frac{\sin \theta}{1 + \cos \theta}$$

$$\tan \frac{\theta}{2} = 2$$

$$\cos \theta = \frac{1-4}{1+4} = \frac{-3}{5}$$

$$\therefore \vec{a} \cdot \vec{b} = \frac{-3}{5}$$

$$\text{Now, } \vec{a} \times \vec{b} = |\vec{a}| |\vec{b}| \sin \theta \cdot \hat{n} = \frac{4}{5} \hat{n}$$

$\vec{c}$ is perpendicular $\vec{a}$ and $\vec{b}$

$\therefore \vec{c}$ is parallel to $\hat{n}$.

$$[\vec{a} \vec{b} \vec{c}] = (\vec{a} \times \vec{b}) \cdot \vec{c} = \frac{4}{5} \hat{n} \cdot \vec{c} = \pm \frac{4}{5} \times 5 = \pm 4.$$

- Q.11 (A) least value of $(a+b+c)$ is 6. (B) $y_1 + z_1$ is equal to -3 .

(D) $\frac{x_3 - x_2}{z_3 - z_2} = 2$

Sol. Planes P_1 and P_2 are perpendicular

$$L: \frac{x+7}{a} = \frac{y-y_1}{-b} = \frac{z-z_1}{c}$$

The above line lies in plane P_2 and perpendicular to plane P_1

$\therefore A(-7, y_1, z_1)$ lies on both the planes.

$$\therefore -14 - 3y_1 + z_1 = -5 \quad \dots(i)$$

$$-7 - 4y_1 - 14z_1 = 5 \quad \dots(ii)$$

Solving (i) and (ii), we get $y_1 = -3$ and $z_1 = 0$

$$\therefore L: \frac{x+7}{2} = \frac{y+3}{-3} = \frac{z-0}{1}$$

Now, verify the options.

Q.12 (A) Q; (B) S; (C) S; (D) P

Sol. $N = 2^2 \cdot 3^2 = 36$.

Now verify the options.

Q.13 (A) T; (B) Q; (C) P; (D) R

Sol.

(A) We have $|\vec{p} + \vec{q} + \vec{r}|^2 = (\vec{p} + \vec{q} + \vec{r}) \cdot (\vec{p} + \vec{q} + \vec{r})$

$$= 1 + 16 + 64 + 2 \sum \vec{p} \cdot \vec{q} = 81 \quad (\text{As, } \sum \vec{p} \cdot \vec{q} = 0)$$

$$\Rightarrow |\vec{p} + \vec{q} + \vec{r}| = 9.$$

(B) Let $A_0(-3, 6, 3)$, $B_0(0, 6, 0)$, $\vec{c} = \langle 2, 3, -2 \rangle$ and $\vec{d} = \langle 2, 2, -1 \rangle$

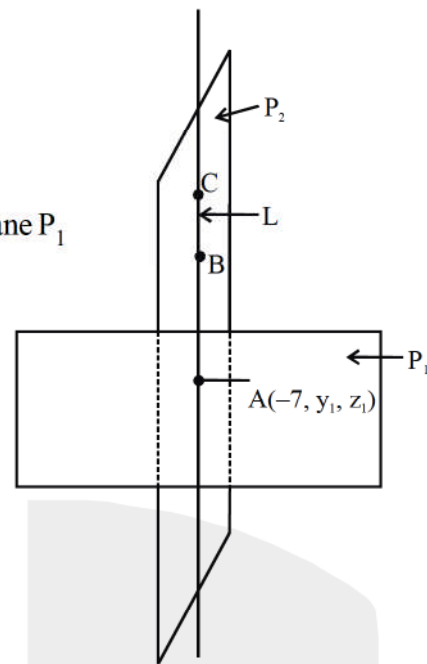
$$\text{Then } AB_{\min} = \left| \text{Projection of } \overrightarrow{A_0B_0} \text{ on } \vec{c} \times \vec{d} \right| = \frac{|\overrightarrow{A_0B_0} \cdot \vec{c} \times \vec{d}|}{|\vec{c} \times \vec{d}|} = \frac{\begin{vmatrix} 3 & 0 & -3 \\ 2 & 3 & -2 \\ 2 & 2 & -1 \end{vmatrix}}{|\hat{i} - 2\hat{j} - 2\hat{k}|} = 3.$$

(C) As, (α, β, γ) lies on plane $x + 2y + z = 4 \Rightarrow \alpha + 2\beta + \gamma = 4$ (1)

$$\text{Now, } \hat{j} \times (\hat{j} \times \vec{v}) = \vec{0} \Rightarrow (\hat{j} \cdot \vec{v})\hat{j} - (\hat{j} \cdot \hat{j})\vec{v} = \vec{0} \Rightarrow (\hat{j} \cdot \vec{v})\hat{j} = \vec{v}$$

$$\Rightarrow \beta\hat{j} = \alpha\hat{i} + \beta\hat{j} + \gamma\hat{k} \Rightarrow \alpha\hat{i} + \gamma\hat{k} = \vec{0}, \text{ which is possible when } \alpha = \gamma = 0.$$

So, from equation (1), we get $2\beta = 4 \Rightarrow \beta = 2$.



(D) Let equation of variable plane be $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} - 1 = 0$

Now, A (a, 0, 0) ; B (0, b, 0) ; C (0, 0, c)

$$\therefore \text{Centroid of tetrahedron OABC} = \left(\frac{a}{4}, \frac{b}{4}, \frac{c}{4}\right)$$

$$\text{So, } x = \frac{a}{4} \Rightarrow a = 4x ; y = \frac{b}{4} \Rightarrow b = 4y \text{ and } z = \frac{c}{4} \Rightarrow c = 4z$$

$$\text{Also, } \frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = 2 \Rightarrow \frac{1}{4} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

$$\text{So, } \frac{16}{4} = \frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2}$$

Hence, $k = 4$.

Q.14 3

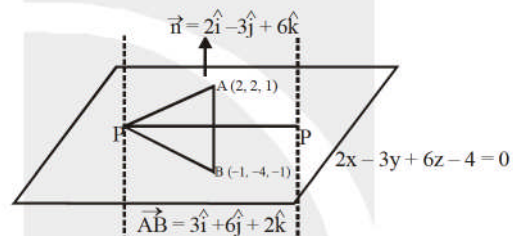
Sol. [Note: A and B lies in the plane π .]

$$\text{Area of } \Delta PAB = \frac{1}{2} (AB) \text{ height} = 14$$

$$\frac{1}{2} \times 7 \times \text{height} = 14$$

$$\text{height} = 4$$

$\therefore$ locus of point P is the pair of straight line parallel to AB



now normal of the required plane i.e. $6\hat{i} + a\hat{j} + b\hat{k}$ is perpendicular to $\vec{AB}$ as well as perpendicular to $\vec{n}$ of the given plane.

$$18 + 6a + 2b = 0 \text{ \& } 12 - 3a + 6b = 0$$

solving we get $a = -2, b = -3$

$$\therefore \text{required plane} = 6x - 2y - 3z + D = 0$$

perpendicular distance from A(2, 2, 1) is 4

$$\left| \frac{12 - 4 - 3 + D}{7} \right| = 4$$

$$\Rightarrow |5 + D| = 28$$

$$D = 23 \text{ or } -33$$

$$\therefore \frac{d_1 - d_2 + a + b}{17} = \frac{d_1 - d_2 + a + b}{17} = 3$$

Q.2 0024

$$\text{Sol. The normal vector of plane } \Pi_1, \text{ is } = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{vmatrix} = \hat{i} - \hat{j} + 2\hat{k}$$

$\therefore$ Equation of plane Π_1 is, $1(x-2) - 1(y-3) + 2(z-4) = 0$ or $\Pi_1 : x - y + 2z - 7 = 0$.

Also, $\Pi_2 : x - y + 2z - 19 = 0$

$$\text{So, } d = \frac{|-7+19|}{\sqrt{6}} = \frac{12}{\sqrt{6}} = 2\sqrt{6}$$

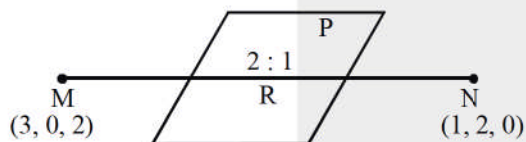
Hence, $d^2 = 24$.

Q.3 7

Sol. Let $P_1 : x + y + z + 3 = 0$; $P_2 : x - y + z - 2 = 0$

Let $P : P_1 + \lambda P_2 = 0$

$$\Rightarrow (x + y + z + 3) + \lambda(x - y + z - 2) = 0 \quad \dots\dots(1)$$



Put $(x = 1, y = 2, z = 0)$ in equation (1), we get $\lambda = 2$

$$\therefore P : 3x - y + 3z = 1$$

$$\Rightarrow (a + b + c) = 3 + 3 + 1 = 7$$

Q.4 1710

Sol. Let equation of a plane containing the line be $l(x-1) + m(y+2) + nz = 0$
then $2l - 3m + 5n = 0$ and $l - m + n = 0$

$$\therefore \frac{l}{2} = \frac{m}{3} = \frac{n}{1}$$

$\therefore$ the plane is $2(x-1) + 3(y+2) + z = 0$

$$\text{i.e. } 2x + 3y + z + 4 = 0$$

$$\therefore a = 2, b = -3, c = 1$$

$$\text{Hence, } (10^3a + 10^2b + 10c) = 2000 - 300 + 10 = 17107$$

Q.5 0001

$$\text{Sol. } \vec{r} = (\vec{a} \times \vec{b}) \sin x + (\vec{b} \times \vec{c}) \cos y + 2(\vec{c} \times \vec{a})$$

$$\vec{r} \cdot (\vec{a} + \vec{b} + \vec{c}) = 0 \Rightarrow [\vec{a} \vec{b} \vec{c}] (\sin x + \cos y + 2) = 0$$

$$[\vec{a} \vec{b} \vec{c}] \neq 0 \Rightarrow \sin x + \cos y = -2$$

this is possible only when $\sin x = -1$ and $\cos y = -1$

for $x^2 + y^2$ to be minimum $x = \frac{-\pi}{2}$ and $y = \pi$.

$$\Rightarrow \text{Minimum value of } (x^2 + y^2) \text{ is } = \frac{\pi^2}{4} + \pi^2 = \frac{5\pi^2}{4}.$$

$$\Rightarrow \tan^{-1} \left(\tan \frac{\alpha}{\pi} \right) = \tan^{-1} \left(\tan \frac{5\pi}{4} \right) = \frac{\pi}{4}.$$